MTH 331(A) Dr. James Ashe
Exam 3 Monday 4th November, 2013 **Name:**

Unless otherwise indicated, each question is of equal value [4 pts]. Show all of your work.

Use $\mathbf{u} = \langle 2, 4, 1\rangle$ *and* $\mathbf{v} = \langle 1, -3, 2\rangle$ *to perform the indicated operations in problems 1–3.*

1. $\mathbf{u} - 2\mathbf{v}$

2. $\mathbf{u} \cdot \mathbf{v}$

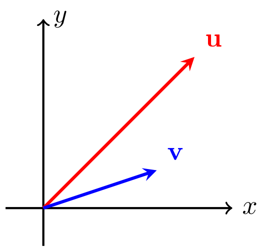
3. $\mathbf{u} \times \mathbf{v}$

Use the graph to draw the following vectors in problems 4–5 (provide labels). Make certain to indicate direction.

4. **a.** $\frac{1}{2}\mathbf{u}$

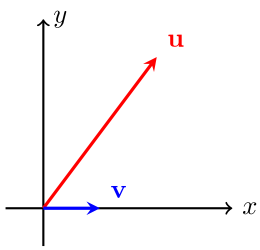
b. $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} - \mathbf{v}$ (you do not have to draw it as a position vector)



5. **a.** $\text{proj}_{\mathbf{v}}\mathbf{u}$

b. $\text{proj}_{\mathbf{u}}\mathbf{v}$

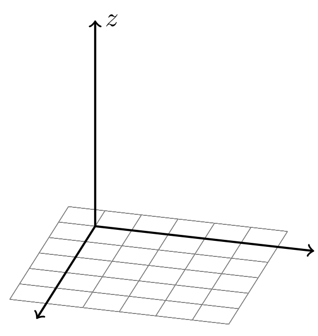


Use the graph to perform the indicated operations (provide labels).

6. **a.** Use the right-hand rule to label the x and y axes.

b. Plot the point $(4, 2, 3)$.

c. Sketch the plane $(x, y, 3)$.



Use the vectors $\mathbf{u} = \langle 3, 6 \rangle$, $\mathbf{v} = \langle 2, 3 \rangle$ *and the point* $P(1, -4)$ *to complete problems 7–10.*

7. Find the line through P and parallel to $\mathbf{u}$.

8. Find the angle between $\mathbf{u}$ and $\mathbf{v}$.

9. Find a vector orthogonal to $\mathbf{u}$ and $\mathbf{v}$.

10. Is the vector found in problem 9 unique? If not find another vector orthogonal to $\mathbf{u}$ and $\mathbf{v}$.

Let $\mathbf{r}(t) = \left\langle 1, 2t^{-2}, e^{-t} \right\rangle$. *Evaluate problems 11–13.*

11. $\mathbf{r}'(t)$

12. $\displaystyle\int_0^1 \mathbf{r}(t)\, dt$

13. $\displaystyle\lim_{t \to \infty} \mathbf{r}(t)$

Compute the slope of the tangent line of the vector valued function at $t = 1$.

14. $\mathbf{r}(t) = \left\langle 1 + t^3, 2\sin 3t, e^t \right\rangle$

Complete problems 15–17.

15. A sailboat floats in a current that flows due west at 3 m/s. The wind is blowing 30 north of east at a speed of 5 m/s. Find the speed of the boat relative to the shore.

16. A constant force vector $\mathbf{F} = \langle 50, 120\rangle$ (pounds) is used to move a heavy object 50 feet along the x-axis. Calculate the work done.

17. A 40-pound weight hangs from the end of a 2-foot bar that is attached to a wall at an angle of 45° below the horizontal.

a. Find the magnitude of the torque about the point on the wall where the bar is attached.

b. Draw a picture and indicate the direction of the torque.